

MUHAMMAD ZAYAB ANSARI

AI Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate specialising in AI engineering and scalable machine learning systems. Experienced in architecting RAG pipelines, agentic AI workflows, and multi-tenant SaaS platforms using FastAPI, LangChain, Docker, and LLMs. Delivered end-to-end AI systems spanning NLP, Computer Vision, and Generative AI from data pipeline design to containerised production deployment with CI/CD and model monitoring. Seeking a junior AI Engineer role to build impactful, scalable AI products.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS

AI / ML: Scikit-Learn, TensorFlow, PyTorch, XGBoost, LightGBM, CNNs, Transfer Learning, Transformers, Fine-tuning, Inference Optimisation

Generative AI: LLMs, OpenAI API, Groq API, Hugging Face, RAG Pipelines, Ollama, Embeddings

Agentic AI: LangChain, LangGraph, FAISS, ChromaDB, Sentence Transformers, Vector Databases, Semantic Search

Computer Vision: YOLOv11, ONNX, OpenCV, Data Augmentation, MRI Analysis, Real-Time Inference

Backend & APIs: FastAPI, REST API Design, SQLAlchemy, PostgreSQL, MongoDB, Redis, JWT Auth, Docker, Nginx, WebSocket, Microservices

MLOps: CI/CD (GitHub Actions), Model Monitoring, MLflow, Containerisation, Hugging Face Spaces, AWS/GCP

Frontend: Next.js, React 18, TypeScript, Tailwind CSS, Streamlit

Tools: Git/GitHub, Jupyter, VS Code, Linux/Bash, Postman

EXPERIENCE

AI Engineering Intern | SMIT Bootcamp 5.0

Jan 2025 – Present

- Engineered and deployed 10+ AI models across NLP, Computer Vision, and Generative AI; containerised via Docker to Hugging Face Spaces, reducing deployment time by ~60% using CI/CD pipelines.
- Integrated LLM-based agentic AI solutions with FastAPI backends; enforced REST API versioning and automated testing via GitHub Actions, cutting manual QA effort by 40%.
- Applied ONNX export and quantisation, reducing average model inference latency by 35% across Computer Vision models.

Freelance AI Engineer | Fiverr (Self-Employed)

Mar 2024 – Present

- Delivered end-to-end ML, NLP/RAG, and Computer Vision solutions for international clients across healthcare, legal, and e-commerce verticals; maintained 5-star rating across all engagements.
- Built and deployed multi-tenant AI SaaS platforms and RAG chatbots on Docker-based cloud infrastructure; reduced project delivery time by 30% via reusable ML pipeline templates.

PROJECTS

Raabta — Multi-Tenant WhatsApp AI SaaS — FastAPI, Groq/Llama 3.3-70B, LangChain, MongoDB, WhatsApp Cloud API
2026

- Architected a multi-tenant WhatsApp AI agent platform for Pakistani SMEs (restaurants, clinics, salons) with per-tenant RAG, JWT auth, and WhatsApp Cloud API integration.
- Load-tested to 150 concurrent users; all production features verified across 4 independent test suites.

Khata — AI Bookkeeping Assistant — FastAPI, Groq/Llama 3.3-70B, MongoDB, React, Recharts

2026

- Built vision-based invoice extraction via Groq Llama with strict JSON schema validation, confidence scoring, and a self-correction loop; 15/15 logic tests passed.
- Delivered spend insights dashboard, MongoDB ledger, Excel/PDF export, and light/dark mode; deployment config via Dockerfile, render.yaml, and vercel.json.

Sentrix — Enterprise AI Security Platform — *Python, FastAPI, XGBoost, LSTM, Docker, React* 2025 – 2026

- 7-model AI ensemble (XGBoost, Isolation Forest, LSTM) for real-time SOC threat detection mapped to MITRE ATT&CK; 300+ automated tests with >98% pipeline reliability.
- Scalable microservices with JWT/RBAC, async PostgreSQL, WebSocket alerting, and 10 automated SOAR playbooks reducing MTTR by design.

Pakistan LegalAI — Final Year Project — *FastAPI, React 18, LangChain, FAISS, LLaMA 3-70B, Groq, PostgreSQL, Docker* 2024 – 2025

- RAG pipeline over 12+ Pakistani laws and 21 court judgments; 95%+ retrieval relevance, <3s average response latency via LLaMA 3-70B on Groq API.
- 25+ JWT-authenticated REST endpoints, Urdu/English bilingual chatbot, legal PDF generation, Know Your Rights module, Docker Compose + GitHub Actions CI/CD.

ClinIQ — Clinical AI Platform — *FastAPI, Streamlit, Random Forest, CNN/ONNX, LLaMA 3.3-70B, Hugging Face Spaces* 2024

- Multimodal platform integrating 3 independent ML microservices (Random Forest, CNN/ONNX, LLM) with <2s diagnostic query response; real-time model monitoring on Hugging Face Spaces.

Brain Tumor Detection & Flight Delay Prediction — *YOLOv11/ONNX, XGBoost, MLflow, Scikit-Learn* 2024

- YOLOv11 transfer learning: 93.5% mAP at 30 FPS on MRI streams, exported to ONNX for cross-platform deployment.
- XGBoost flight delay predictor: 89% precision/recall, all experiments tracked via MLflow.

EDUCATION

B.E. Software Engineering 2022 – 2026

Final Year Project: Pakistan LegalAI Platform — production-ready RAG system with 95%+ retrieval accuracy; deployed via Docker + Nginx with full CI/CD pipeline.

Coursework: Big Data Analytics (Hadoop, Spark), Machine Learning, Database Systems, Computer Networks, OOP, System Design.

Data Science & AI Bootcamp 2025 – 2026

Focus: Deep Learning (CNNs, Transformers, Transfer Learning), LLMs, RAG Pipelines, Agentic AI Workflows, Model Fine-Tuning, MLOps Fundamentals.

CERTIFICATIONS & ACHIEVEMENTS

- Cisco Networking Academy: Python Essentials 1 (Dec 2025) & Python Essentials 2 (2026)
- Kaggle Certifications: Computer Vision, Deep Learning, Machine Learning, Data Cleaning, Data Visualization, Python
- Hackathons: Coding Night 2026 Hackathon Participant